TaMoS- Answer- Collection

Vetenskapsteori och vetenskaplig metodik (naturvetenskap) (Kungliga Tekniska Högskolan)



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TaMoS AK2030 MCQ Answers

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Which of the following claims about observations are true?

- Strictly speaking if you are using glasses or contact lenses you are doing aided observations.
- In successful indirect observations you are observing an event causally connected to the feature of interest, rather than the feature of interest itself.
- Direct observations can be aided

In what ways do qualitative and quantitative data differ?

- Large quantitative data sets are easier to summarize than large qualitative data sets.
- Qualitative data is rarely measurable on interval or ratio scales, and therefore there is often no need to specify a unit. To produce quantitative data (measured on interval or ratio scales), a specification of a unit for comparison is required
- When collecting quantitative data, one typically focuses on a few properties, qualitative data on the other hand captures a broad range of properties.
- Scientists often use different methods to produce qualitative and quantitative data, respectively

What is true about utilitarianism? According to utilitarians...

- … saving ten people by killing five people could be morally right, everything else being equal.
- … the morally valuable property is utility, usually characterized as happiness.
- … utility, often interpreted as happiness in some form, is the sole good.

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What is true regarding the virtues of models?

- When comparing two models, valuing precision means choosing the one whose parameter specifications imply the parameter specifications of the other.
- Having the virtue theoretical tractability means that the model can, in a scientifically meaningful way, be processed by a scientific theory.
- When a model user can understand how the model produces the result, then it is epistemically transparent.
- In changing one of the assumptions in a model does not change the model result, then the model is robust with respect to this assumption.

What are necessary conditions for a good operationalization?

- The observable property and the property of interest must be connected in a sufficiently stable way.
- The observable property and the property of interest must be connected in a valid way, to our best knowledge.
- Given the practical context, the observation of the observable property must be sufficiently precise.
- The connection between a feature of interest and a directly observable property must be sufficiently stable for practical purposes. / valid according to our best knowledge.
- The observable property must be observable in practice with sufficient precision.
- The operationalization must connect the feature of interest with an observable property.
- It must connect the property of interest with an observable property
- The observable property must be observable in practice with sufficient precision

What is true regarding the steps taken in the methodology of design?

- One step - validation - is showing that it fulfills the functional requirements.
- The design needs to be validated, which means to show that it fulfils the functional requirements.
- An important step is identifying possible physical descriptions, for instance by considering operational principles and normal configuration
- When deciding between possibilities, you can either aim for optimization (doing an exhaustive search) or satisficing (determining a minimal threshold)
- The methodology of design consists of the same steps as the methodology of natural science.

One part of the classical definition of knowledge concerns belief. What is true regarding the concept of belief?

- Beliefs can be more or less justified.
- A person can believe something that is false.
- Beliefs are personal - they are something that a person has.
- It is possible to believe something that is not true.
- Beliefs are action guiding in sense that actions depend on beliefs.
-

What is true of learning from models as analogies?

- By referencing the positive analogies, we can justify the use of neutral analogies.
- Knowledge of a negative analogy is when we know that something is different between the model and the target

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What is true about falsifiable statements? A falsifiable statement ...

- ... might be false.
- ... might be true.
- ... can be about something that is not directly observable but will still be testable.
- ... is either true or false, but not both.
- … is neither true or false.
- … are neither necessarily true nor necessarily false.
- ... are testable even though they may be about something not directly observable.

What is true regarding explanatory virtues (what makes explanations powerful)?

- Being less sensitive to background causes than another explanation means that the explanation is more powerful than the other.
- The contributing causes that are underlying the explanandum should be identified in the explanans.
- An explanans should identify those contributing causes that produce the difference asked for in the contrast of the explanandum.

What is an ad hoc hypothesis?

- A consequence of ad hoc modifications is a decrease in falsifiability.
- It is a change C of a theory T resulting in an updated version T' such that C attempts to include some observations so that T is saved from being falsified.
- It is a modification of a theory that attempts to accommodate some unexpected observations only in order to save the theory.
- It is a modification of a theory that decreases falsifiability.
- It is a modification of a theory that does not propose a new means of testing the theory.
- It is a kind of claim that shares some (or all) observable consequences with the previous hypothesis.

Which of the following claims about experiments and observational studies are correct?

- In an observational study, one does not intervene.
- One can perform an experiment in a natural environment.
- Inferences drawn from experiments and observational studies can be equally justified.
- Inferences from observational studies might be as internally valid as from experiments.
- Inferences from observational studies might be as good as those from experiments.
- Both observational studies and experiments can take place outdoors in a natural environment.
- Experiments include control and manipulation.
- Observational studies do not include an intervention.

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What is true about episteme and techne?

- Techne provides the ability of achieving some practical goals.
- Episteme means theoretical knowledge, techne means practical knowledge.
- Techne has deep and complex methodological problems of its own.

Which of the following claims about words and definitions is/are true?

- In a lexical definition of a value-laden term, the term's value-ladenness should be accurately reflected.
- A stipulative definition is not necessarily narrower than its lexical counterpart.
- A vague word lacks a sharp limit between what it applies to and what it does not apply to

Which of the following is observable only indirectly?

- The charge in a battery.
- The well-being of a child.
- The pH value of a liquid.
- The precise temperature of a liquid

Which of the following expresses a value on a ratio scale?

- Usain Bolt set a world record, running 100 m in 9.58 seconds.
- A ship travels at 20 nautical miles per hour.
- The distance between the sun and the earth is on average eight light minutes

What is correct about the repeatability of experiments?

- An experiment is repeatable if other competent persons that follow the given description of the experiment can repeat it.
- Repeatability does not imply replicability.
- The requirement of repeatability does not mean that scientific experiments actually have to be repeated, only that it should be possible to repeat them given the published description.
- Repeatability of an experiment can be determined by analyzing the published material about this experiment.

What is correct about the peer review system? Select all true claims.

- The editor of a scientific journal sends off incoming articles to experts in the area and bases the decision whether to publish or not on their judgment.
- It is a system that lets scientists from the same field of research as the author of an article, so called "peer reviewers", critically evaluate the scientific work of that article.
- It is a system created to uphold scientific quality of the submitted paper, especially regarding the novelty and justification of its claims.

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Scientific openness is generally valued in science, however there are legitimate exceptions. Not sharing scientific data may be justified when . . .

- . . . sharing the data jeopardizes the chances of publishing the results in an article.
- . . . protecting people's privacy.
- . . . the correctness of the data has not been checked.
- . . . protecting intellectual property.

What is true about the precautionary principle? The precautionary principle . . .

- . . . states that the proponent of an activity, rather than the public, should bear the burden of proof of the activity's harmlessness.
- . . . implies that the scientist must be able to show that the new detergent is not harmful, or that the new species does not harm the ecosystem, etc.
- . . . imposes a moral constraint on researchers.
- . . . states that if an action or policy has a suspected risk of causing harm to the public or to the environment, in the absence of scientific consensus that the action or policy is harmful, the burden of proof that it is not harmful falls on those taking an action.
- . . . states that when an activity raises threats of harm to human health or the environment, precautionary measures should be taken even if some cause-and-effect relationships are not fully established scientifically.

What is true about the precautionary principle?

- implies that if the scientist A creates a new detergent, then scientist A must be able to show that the detergent is not harmful.
- imposes a moral constraint on researchers.
- states that when an activity raises threats of harm to human health or the environment, precautionary measures should be taken even of some cause and effect relationships are not fully established scientifically.
- states that if an action or policy has a suspected risk of causing harm to the public or to the environment, in the absence of scientific consensus that the action or policy is harmful, the burden of proof that it is not harmful falls on those taking an action

What is true about knowledge according to the classical definition of knowledge? According to the classical definition of knowledge . . .

- . . . if you know that it is raining then you are justified in believing that it is raining.
- . . . if it is not raining at place P at time t then you do not know that it is raining at place P at time t.
- . . . if you know that it is raining at place P at time t, then it is raining at place P at time t.
- . . . a person might know P even though she believes P with less than certainty.
- . . . beliefs and justifications are infallible.

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Does methodology aim to eliminate all errors in science?

- No. Available evidence often gives us reason to choose one method over the other, but these reasons are often neither complete nor certain.
- No. One argument for this is that science cannot eliminate all errors, only manage potential sources of error.
- No. Almost all our beliefs are fallible, so it would be impossible to completely eliminate error.

Does methodology require scientists to completely eliminate the risk of error in science?

- No, science involves a trade-off between the utility of having knowledge, and the risk of being wrong.
- No, one argument for this is that science cannot eliminate all errors, only manage potential sources of error.
- No, almost all empirical claims are fallible, so an empirical science that aims to completely eliminate error would lose much of its content.
- No. Available evidence often gives us reason to choose one method over the other but these reasons are often neither complete nor certain.

What distinguishes valid deductive inferences from inductive inferences?

- Valid deduction is truth-preserving, justified induction is fallible.
- Generalization is an example of inductive inferences, modus tollens is an example of deductive inferences.
- Inductions extend conclusions beyond knowledge we already have; deductions order and rearrange our knowledge without adding to its content.
- Projection is an example of inductive inference, modus ponens an example of deductive inferences.

What is correct about the hypothetico-deductive method for hypothesis testing?

- Operationalization of not directly observable features of interest is necessary for the hypothetico-deductive method.
- Observable consequences are deduced from one's hypotheses and tested empirically.
- If the outcome of a test (i.e. an experiment) is negative, this does not necessarily mean that the hypothesis should be rejected. The negative result might be explained by an auxiliary hypothesis being false.
- If your hypothesis contains non directly observable features, their operationalization becomes a necessary part of the hypothetico deductive method.
- Hypotheses are tested through observable consequences.

- What is true regarding the problems with falsificationism and the hypothetico deductive method?
- The Duhem/Quine thesis poses potential problems for falsificationism.
- In actual scientific practice, hypotheses have confidences, some are considered more likely to be true.

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What is true about "Hume's Problem of Induction"-argument? Hume argued . . .

- . . . that there is no necessary connection between past and future observations.
- . . . that in our inferences, we rely on inference rules that themselves require justification.
- that induction cannot be justified by foundational principles.
- . . . that the use of inference rules in induction cannot be justified by past successes of applying such a rule.

What kind of measurement scale is employed when one ranks movies on the scale 1,2,3,4,5?

- A scale that represents less information than a ratio scale.
- The kind of scale that is also used when describing the final positions in a marathon.
- A scale that represents less information than an interval scale.
- An ordinal scale.

Which of the following claims about systematic error is correct?

- Systematic errors are (in principle) predictable and deviations from the true value are typically constant or proportional to the true value.
- If a cause of the systematic error is identified, then it can usually be corrected.
- Reducing systematic errors increases accuracy, but not precision.
- (Experimental) blinding is a strategy to reduce systematic error.

Which of the following claims about random error is correct? Select all true claims.

- Random errors are inherently unpredictable, and have null expected value
- Reducing random errors increases precision, but not accuracy
- Random errors can be reduced by trying an alternative operationalization
- Random errors can be reduced by repeating measurements and averaging over outcomes

Which factors would threaten the internal validity of inferences from an experiment (in contrast to an observational study)?

- Misinterpreting experimental observations.
- Misunderstanding the influence of experimental manipulation.
- Failing to (materially) control background conditions.
- Misunderstanding the influence of background conditions.
- ~~Failing to justify why an experimental result applies to a relevant real-world situation.~~

What are good (epistemic or non-epistemic) reasons why scientists build and investigate models of targets they are interested in, rather than investigating these targets directly?

- It is too expensive to handle some targets directly.
- Some targets are too complex to gain knowledge from their direct investigation.
- Ethical or political reasons prohibit the direct manipulation of some targets.
- Some targets cannot be handled directly for practical reasons.

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What is true about valid isolating models?

- A valid isolating model is dissimilar to its target and still licenses relevant (externally) valid inferences.
- A valid isolating model singles out one aspect of the target, ignoring all the others.
- A valid isolating model represents the workings of the isolated factor accurately.

Why should hypotheses be tested statistically?

- Statistical accounts often impose weaker demands on hypothesis testing than (non-statistical) accounts of confirmation and falsification.
- Using probability calculus is a way to model and calculate the degree of confirmation that some evidence confers to a hypothesis.
- Some hypotheses only make claims about distributions of outcomes.
- Observations contain random measurement errors. A single (seemingly) falsifying observation might thus not be a good reason to reject a hypothesis.

What is true of the Deductive-Nomological (DN) Account of Explanation?

- The DN account is problematic, because some cases satisfy the DN account's conditions, but intuitively do not constitute genuine explanations.
- According to the DN account, understanding of a phenomenon is achieved by having reasons for why the phenomenon was expected on a lawful basis.
- According to the DN account, the logical structure of scientific explanations and predictions is very similar.
- According to the DN account, we achieve genuine understanding of a phenomenon by identifying laws that give reasons to expect the phenomenon to occur.
- According to the DN account, an explanation consists in the deduction of the explanandum from laws of nature and initial conditions

What is true about informed consent? Informed consent . . .

- . . . is not something that everyone can give.

- . . . designates the legitimate authorization of research participation.

- . . . is usually understood as having three conditions that need to be fulfilled.

. . . cannot be given involuntarily.

What is realism in the theory of science?

- . . . for example the view that electrons exist, even though they are not directly observable.
- is the view that there is an existing reality independent of us and our senses.
- the view that the scientific theories describe, or aim at describing, what the world is really like.
- the view that we have reason to believe that our best theories of the world are literally true.

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What is the relationship between scientific methods and methodology?

- Methodology concerns about method choice \ is about method choice.
- Methodology is a way to justify why you do science in a certain way and not in another -methods is how you do it.
- The basis for methodological reasoning is to determine what a proper argument for the selection of a method would be.
- For different methods, the same methodology might apply.
- Methodology critically assesses the application of methods for certain purposes.
- ~~A number of methods in a given scientific discipline can be said to jointly constitute the methodology of that discipline.~~

Which of the following sentences express good hypotheses?

- "Computer program of type A uses more memory than computer program of type B"
- "The Ebola epidemic was caused by bacteria".
- "Individuals with high level of education judge climate change as a high risk".

Which of the following sentences satisfy the minimal criteria for hypotheses?

- "This female gave that student a lower grade because the student was female."
- "Individuals with a high level of education judge climate change as a high risk."
- "Computer program of type A uses more memory than computer program of type B."

Which of the following sentences can be hypotheses?

- "Asphalt of type C1 has longer lifespan than type C2."
- "This researcher disregarded the first trial due to that scientist being biased."
- "People in rural areas have a lower tolerance to viruses."
- "Mercury thermometers do not give accurate results at temperatures close to -37 degrees Celsius."

What is true of the 7 base units of the International System of Units (SI)?

- Some of the units may be more fundamental than others.
- An unlimited number of derived units can be formed from the SI base units through mathematical operations.
- It consists of the units: ampere, kelvin, mole, candela, meter, kilogram and second.

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What is true of Karl Popper's falsificationism?

The falsifictionalist..

- never accepts any hypothesis as true - only maintains non - falsified hypotheses as so far not rejected.
- seeks to falsify hypotheses with observable evidence.
- conjectures (proposes) falsifiable hypotheses.
- proposes falsifiable hypotheses.
- aims to reject (but not accept) hypotheses.

Which of the following claims about disturbing factors in experiments is correct?

- Observer influence may be a source of disturbing factors in an experiment.
- One eliminates a disturbing factor by making sure that the disturbing factor is not present.
- When we cannot eliminate disturbing factors, we try to separate them and register their influence independently.
- Randomization is an important but not necessary tool for dealing with disturbing factors: by eliminating selection bias and also facilitating blinding.

Which advantages do experiments have over other empirical methods for testing hypotheses?

- Causal hypotheses are more easily tested with experiments.
- Experiments offer great control of many relevant background conditions than other empirical methods.

What is the relation between models and experiments?

- Performing experimental control is often a difficult task. Setting variable and parameter values in mathematical models, in contrast, is often trivial.
- Both models and experiments involve manipulations. But in models these are always manipulations of representations.
- Internal validity is generally less of a problem for models than for experiments.
- Models are always representations of targets different from themselves, while experiments are sometimes performed on the target systems themselves.
- External validity is more of a problem for experiments than for models

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Which claims are true of Fisher's Null hypothesis Testing?

- How the significance level is set is a mere convention.
- Fisher's Null Hypothesis Test only helps decide whether to reject the null hypothesis or not.

Which claims are true of Fisher's significance testing?

- How the significance level is set a mere convention.
- Fisher's Significance Test only helps decide whether to reject the hypothesis or not.
- The result of Fisher's Significance Testing can be influenced by changing the experimental design, and by changing the partition of possible outcomes.
- When using Fisher's significance testing, the only conclusion we can be certain of is the rejection of the hypothesis.
- By changing the study design, and by changing the partition of possible outcomes, the result of Fisher's significance testing can be influenced.

What is true about explanans (explanantia) and explanandum (explananda)?

- Explanantia that are not fully accurate (and hence false in that sense) can nevertheless sometimes explain.
- Failing to specify the contrast of an explanandum might lead to multiple correct competing explanations.
- The explanandum is what requires an explanation, the explanans is what provide the explanation.
- The less specific the contrast of an explanation is, the higher the number of explanantia that give correct but competing explanations.
- A singular explanandum can be explained with a singular or general explanans.

What is true of tacit knowledge?

- Tacit knowledge is difficult to transfer to another person by means of writing it down or verbalizing it.
- To know something tacitly means that one has a skill, experience or idea without necessarily being able to describe it explicitly.
- Teaching tacit knowledge typically requires personal contact and interaction.
- Teaching tacit knowledge requires extensive personal contact, regular interaction and thrust.
- The ability to ride a bicycle is an example of tacit knowledge.

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Which of these are theories of normative ethics?

- Consequentialism.
- Deontology.
- Virtue ethics.
- Kantianism.

What is true about how risks are perceived? Psychological studies have shown that…

- involuntary risks are seen as worse than voluntary risks.
- risks where one lacks control are seen as worse.
- risk to identifiable individuals are seen as worse than statistical risks (with same or greater number of expected deaths)
- . . . new risks are seen as worse than old risks.

What is true about risk assessment? Risk assessment is the step in risk analysis where….

- standard scientific techniques are utilized.
- experiments or observational studies are performed.
- Statistical evidence is considered.
- . . . epidemiological studies are performed.

What are some technical definitions of risk? Risk is

- the statistical expected value of unwanted events which may or may not occur.
- the severity of an unwanted event which may or may not occur.

What is true about deontology? Deontologists . . .

- . . . are more inclined than others to think in terms of categorical imperatives

. . . think that actions are right or wrong in themselves.

What is true about inductive inferences?

Inductive inferences….

- are fallible.
- require justification.
- do not guarantee that the conclusions are true.
- have premises.

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What is correct about blinding in experiments?

- You could use double blinding to reduce observer expectation effects (i.e. the interpretation problem).
- One condition for single blinding is that subjects do not know which treatment they are getting.
- One condition for double blinding is that experiments do not know which treatments are administered to what subjects.
- Single blinding is an effective strategy to reduce placebo effects and thus to increase the interval validity of an experiment.
- One condition for single blinding is that subjects do not know if they are receiving treatment or not.

Identify the relevant epistemic virtues of scientific models from the following list.

- Robustness
- Precision
- Simplicity
- Generality
- Transparency

Which of the following claims about statistics are true?

- The application of standard statistical tools to good-quality data can lead to false conclusions.
- Statistics provides a toolbox of descriptive and inferential methods; and statistical reasoning justifies the choice from this toolbox.
- "Lying with statistics" often consists in the intentional or accidental misapplication of valid statistical methods to good-quality data.
- Statistics offers different algorithms that might yield different results when applied to the same kind of data.

Which claims are true about Bayesian statistics?

- In Bayesian statistics, the probability of a hypothesis is changed if new evidence is observed.
- Bayesian statistics can evaluate a potentially large set of competing hypotheses.
- Bayesian statistics interpret probabilities as "degrees of belief" or "subjective confidence inhypotheses".
- The "problem of old evidence" remains a substantial problem for Bayesian statistics.
- In Bayesian statistics, the probability of a hypothesis is changed as a consequence of observing new evidence.

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Which claims are true about the relation between correlation and causation?

- Correlation measures the covariance between variables. Causation measures the (productive) influence of one variable on another.
- Variables can be correlated with each other even though no direct causal relation exists between them.
- Many different causal models between two variables are compatible with the observation that the two variables are correlated.
- Correlation between two variables is symmetric. The causal relation between two variables is asymmetric.
- Two variables can be correlated with each other even though neither one is caused by the other.

Which of the following strategies can help improve the accuracy of interpretation of behaviour in usability experiment?

- Use inter-subjective checks.
- Observing indications of cognitive processes, for example a "thinking out loud protocol".
- Check for coherence by repeating questions in questionnaire or interview.
- Ensure that the observer is competent both with the artefact and with its use.
- Gather evidence for cognitive processes

What is true about lexical definition?

- A good lexical definition is neither too broad nor too narrow.
- A lexical definition can be imprecise.
- A lexical definition attempts to capture how a dictionary would define a term.
- A lexical definition attempts to capture how an expression is normally used.
- A lexical definition attempts to capture the common usage of a term.

What is true of the Duhem-Quine thesis? According to the Duhem-Quine thesis...

- we cannot test our theories claim by claim.
- we never test a single hypothesis alone.
- to identify what the falsifying observation falsifies requires judgement and is not settled by deductive logic alone.

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Which of these is a way in which Neyman-Pearson Decision Theory and Fisher's Significance testing are similar?

- A significant result in the test tells us nothing about the probability of the null hypothesis given the evidence.
- Nothing about the probability of the hypotheses is implied by a non-significant result.
- They take into account the probability of making the observation given that the hypothesis is false.
- The probability for the outcome given a hypothesis is compared to a threshold level.
- The tests say nothing about the probability of the hypothesis.
- Observations might contain random measurement errors. A single (seemingly) falsifying observation might thus not be a good reason to reject a hypothesis.

Which of the following are type explanations?

- Most house fires are due to unattended candles.
- Birds can follow their migration routes because they have receptors in their eyes that are sensitive to the magnetic field of the earth.
- Roads that repeatedly go from double to single file and back again have higher accident rates because drivers are bad at judging overtaking maneuvers.

Which claims are true either of methodological individualism or holism?

- Methodological holism claims that there are good social explanations that do not provide individualist accounts.
- Methodological holism claims that there are good social explanations that do not provide individualist accounts.
- Methodological holism is compatible with the claims that all social phenomena are produced by individual interactions.
- Methodological individualism claims that all good social explanations provide individualist accounts of the explanandum
- Methodological holism claims that sometimes, individualist accounts do not provide illuminating explanations of social phenomena because they can be instantiated in many different ways on the individual level

What are necessary steps in a measurement procedure?

- Representation by a measurement scale.
- Specification of a measure and definition of a unit.
- Specification of measurement operation(s).
- Definition of the concept being measured.

What are necessary aspects of scientific experiments?

- The intervention.
- The manipulation of some real variable.
- An observation of the result of manipulation.
- The control of background variables.
- That the intervention is not done on a representation of a variable, but instead a variable that is present.

What is true about justification of method choice?

- Scientists are sometimes not justified in their method choices.
- The practice of justifying method choice is called methodology.

What is said to be epistemic goals of science?

- Identifying the cause (or causes) which produced an event.
- Knowing that an artefact satisfies a certain function.

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What is true about interval scales?

- Interval scales are identical up to positive linear transformations
- On (linear) interval scales, arithmetic means are meaningful as averages.
- One can meaningfully speak of how many times greater or lesser the difference between two points on the scale is than the difference between another pair of points
- Time measured by calendar years (such as "2015 A.D." or "500 B.C.") is an interval scale

In the social sciences, how can the quality of an interpretation of human action be improved?

- Generate additional evidence for the assumed cognitive processes, e.g. through eye tracking or thinking aloud protocols.
- Check for coherence between answers, e.g. through (slightly reformulated) repeated questions.
- Use intersubjective checks of the interpretation, e.g. by involving multiple researchers in the analysis.

What is true about Mill's method of difference?

- According to Mill's method of difference, we can determine causal relations by performing experiments. / It states that we can only determine correlations by performing experiments.
- It requires successful control and intervention.
- By activating a factor in one system and not activating in in another.

What is true about Randomized Control Trials (RCTs)?

- Stratification can help with disturbing known background factors equally between test and control group.
- Subjects are separated into test and control by a random process.
- RCTs facilitates both single and double blinding.
- It does not guarantee that background factors are equally distributed between test and control in a test.

What distinguishes models from theories?

- In those scientific domains where models and theories co-exist (e.g. chemistry), theories typically express general principles, which are applied in the building of models.
- The main reason to reject a theory is that it is false. The main reason to reject a model is that it is not useful for a given purpose.
- Scientists who use a theory have reasons to believe that it is true, while scientists who use a model typically know that it is false in several respects.
- By proposing a theory, one claims that the theory's target is indeed governed by the principles expressed in the theory. When using a model, one is typically aware of alternative representations that are useful for other purposes
- Theories make general claims, models can only represent particulars

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What is true regarding measurement error?

- If a measurement unit is based on a prototype that change over time, the measurement stability will be threatened.
- If a measurement is made several times and gives similar values, then the measurement is precise.
- A measurement can be both inaccurate and imprecise at the same time - if the operationalization is affected by both systematic and random error.

What is true about belief-desire explanations? Belief-desire explanations

- can be used to explain how aims of individuals influence their behaviour.
- can be used to explain any system's or organism's actions.
- explain human behaviour as a result of mental states.
- are often used to explain human behaviour.

What are the main claims of anti-naturalist views of the social sciences?

- The social sciences, in contrast to the natural sciences, focus on individual cases and cannot generalize from them.
- The social sciences are fundamentally different from the natural sciences, for instance because the social sciences work with qualitative data and case studies which the natural sciences do not.
- The social sciences cannot and should not follow the methodology of the natural sciences.
- The social sciences, in contrast to the natural sciences, are in the business of interpreting human behavior. Interpretation does not produce causal explanations, but some other form of understanding.
- The social sciences, in contrast to the natural sciences, derive lawlike generalizations from case studies and qualitative data.

What is true regarding the relationship between "design" and "science"?

- The goals of science are explanation and prediction, whereas the goal of design is the satisfaction of functions.

Which of these are correct description rules used to manage uncertainty whose probabilities cannot be quantified?

- "MaxiMin": We should identify the option with the least bad outcome and choose that option.
- "The precautionary principle": the proponent of a possibly harmful activity bears the burden of providing that it is harmless.

What is a stipulative definition?

- A stipulative def. states how the author herself intends to use a word of expression and thus how it is to be understood in the given context which might differ from common usage of the term.

What is the logical empiricists (logical positivists) view of the relation of theory and observation?

- It is the view that theoretical sentences are inferred (or fully justified) from observational sentences.

What is the distinction between operationalization and operationalism?

- Making an operationalization is to provide way of linking a feature of interest to observable properties. Operationalism is the view that even most basic concepts are defined through the operations by which we measure them.

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Define “direct cause”: X is a direct cause of Y..” and provide an example of such a cause.

- Definition: X is a direct cause of Y with respect to a background variable set V if there is a possible intervention on X that will change Y when all other variables in V are held fixed.
- Example: For instance, a billiard ball, A, collides with another billiard ball, B, making B move.

What is the distinction between a direct and contributing cause?

- Changes to contributing causes do not on their own guarantee changes to their indirect effects. The indirect effects spring from chains of direct causes from the contributing cause. Whereas changes to direct cases on their own are sufficient to bring about changes to their effects, keeping the background variables constant.

What is true regarding the two kinds of justification?

- Hume’s problem of induction is not a problem for a coherentist.
- Foundationalism provides justification by identifying basic claims from which the claims to be justified can be inferred.
- Coherentism provides justification consisting of a non-contradictory system with accepted claims, and any claim is justified if the claim can be incorporated in the system.
- Foundationalism suggests that a claim is justified only if it is inferred from a set of self evident axioms.
- Coherentism favour justification consisting of a coherent system with accepted claim, and any claim is justified if the claim can be incorporated in the system.

What is true regarding theory-dependence?

- Some theories help making observations, which in turn justify these theories — this is called the experimenter’s regress.

Consider the following descriptions of three treatments:

(A)“A medical intervention results in a 33% relative decrease in the incidence of fatal myocardial infarction”.
(B)“A medical intervention results in a 2.6 %-point decrease in the incidence of fatal myocardial infarction – from 5.3%’ to 2.7% ”.
(C)“91 persons must be treated with the intervention to reduce the number of fatal myocardial infarction from 3 to 2”.
What is true of these descriptions?

- (C) represents the risk reduction in natural frequencies.
- Any intervention correctly described by (C) is also correctly described by (A).
- (B) represents an absolute risk reduction.
- It has been shown that most people consider treatments represented in format (A) more effective than treatments represented in format (C).

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A decision has to be made between four alternative courses of action.
On alternative A, there are two possible outcomes: 100 people dying (probability 0.6) or no one dying (probability = 0.4). On alternative B, there are four possible outcomes (equal probabilities): 20 people dying, 100 people dying, 400 people dying, or 4 people dying. On C, there are two possible outcomes: 800 people dying (probability 0.01) or no one dying (probability 0.99). On D, finally, there are two possible outcomes (equal probabilities): 30 people dying, or no one dying.
Given the expected value criterion for decisions involving risks:

- C is the best alternative.
- B is the fourth best alternative.
- A is the third best
- D is the second best alternative

A decision must be made between four alternative courses of action. On alternative A1, there are three possible outcomes: 20 people dying (probability 0.2), 500 people dying (p = 0.002), or 5 people dying (p = 0.798). On alternative A2, there are only two possible outcomes (identical probabilities): 20 people dying, or no one dying. On A3, there are three possible outcomes (identical probabilities): 9 people dying, 4 people dying, or 12 people dying. On A4, finally, there is a one percent probability of 800 people dying (otherwise, no one dies).
Select all correct statement(s), given the expected value criterion for decisions involving risks:

- A2 is the fourth best alternative.
- A3 is the second-best alternative.
- A4 is the best alternative.
- A1 is the third best alternative.

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